



DURBAN UNIVERSITY OF TECHNOLOGY
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FACULTY OF ACCOUNTING AND INFORMATICS

DEPARTMENT OF INFORMATION TECHNOLOGY

2024

MID YEAR MAIN EXAMINATION

INSTRUCTIONAL PROGRAMME : ADVANCE DIPLOMA ICT

INSTRUCTIONAL OFFERING : Research Skills

PAPER NUMBER : 1

SUBJECT CODE/S : RESK401

DATE : 19 JUNE 2024

DURATION : 3 HOURS

TIME : 09H00 - 12H00

TOTAL MARKS : 100

NUMBER OF PAGES : 4

EXAMINER/S : MISS LS MANQELE

MODERATOR/S : DR B MUTANGA

INSTRUCTIONS/REQUIREMENTS:-

1. This examination comprises of FOUR questions sections .
2. For QUESTION 1 AND QUESTION 2, use MCQ cards.
3. Use your pen or pencil. Make sure you write your Surname and initials with student number on the MCQ card
3. Use the mark allocations as a guide for the length of your answers.
4. On your MCQ cards: Question 2 asks if the statement is true or false. A is true and B is False. Start question 2 from number 16 on your MCQ cards
5. Attempt ALL QUESTIONS.

Do not turn the page until permission is given

Question 1 [30]

1. This type of research aims to look at how things are now and not pay attention to any trends. [2]
 - a. Explanatory research
 - b. Descriptive research
 - c. Cross-sectional research
 - d. Longitudinal research
2. The following statement is true about the research problem [2]
 - a. The research problem can be replaced by the hypothesis
 - b. The research problem answers "why or how" questions
 - c. The research problem answers "what" questions
 - d. Is the statement that identifies what you intend to investigate
3. The relevancy of the source can be gauged by: [2]
 - a. Abstract
 - b. The research aims
 - c. The findings
 - d. Problem statement
4. Why is referencing so important? [2]
 - a. To discredit the original author's credit for their ideas and work
 - b. To validate your arguments, statements and claims
 - c. To paraphrase the statement and express ideas and thoughts
 - d. To use an idea of another researcher
5. This type of research focuses on testing theories and hypotheses [2]
 - a. Qualitative research
 - b. Mixed research
 - c. Quantitative research
 - d. None of the above
6. The type of survey that can be used to show how employees' opinions about working from home changed from the COVID-19 period up to date. [2]
 - a. Cross-sectional survey
 - b. Panel survey
 - c. Cohort Survey
 - d. Trend survey
7. A research strategy that does not rely completely on the collection and analysis of data. Instead, the focus is on practical research [2]
 - a. Survey

- b. Design Research
 - c. Experiment
 - d. Case study
8. A research strategy that designs and distributes questionnaires to participants [2]
- a. Survey
 - b. Design Research
 - c. Experiment
 - d. Case study
9. This is the type of sampling that is useful when the research requires unique characteristics of the sample and there are few available or difficult to reach participants. [2]
- a. Expert sampling
 - b. Snowball sampling
 - c. Purposive sampling
 - d. Cluster sampling
10. The type of sampling that is selected according to some pre-determined standard. [2]
- a. Quota sampling
 - b. Snowball sampling
 - c. Purposive sampling
 - d. Stratified sampling
11. Type of data collection that highlights trends, relationships and associations between two or more variables. [2]
- a. Qualitative data
 - b. Descriptive data
 - c. Quantitative data
 - d. Relational data
12. This type of focus group consists of two moderators who deliberately take opposite sides on the issue under discussion. [2]
- a. Dual moderator
 - b. Duelling moderator
 - c. Correspondent moderator
 - d. Client participant
13. One focus group watches another focus group and discusses the observed interactions and conclusion [2]
- a. Dual moderator
 - b. Duelling moderator
 - c. Two way
 - d. Correspondent moderator

14. In data collection, the level of agreement can be “disagree, neutral or agree”, this type of data collection is called: [2]

- a. Nominal
- b. Interval
- c. Ratio
- d. Ordinal

15. A cellphone number can be categorized as: [2]

- a. Personal data
- b. Sensitive data
- c. Not personal data
- d. All of the above

Question 2 [20]

1. Basic research aims to solve a practical problem faced by society, industry or some other significant group. [2]
2. Hypothesis may be posed instead of research questions. [2]
3. Elsevier is an example of a publisher that is recognized by international quality standards bodies. [2]
4. Referencing is a means of providing evidence and supporting claims made in the write-up to only deserving authors. [2]
5. Data collection techniques of the mixed method may include interviews, case studies, expert opinions, focus groups, open-ended surveys and observation. [2]
6. A cross-sectional survey is used to collect data relevant to that for not more than seven years. [2]
7. Probability sampling is the most precise type of sampling, every member of the population has an equal chance of being selected. [2]
8. Descriptive data allows for a comparison between two or more sample groups based on one or more variables. [2]
9. Interval data has a ranking and the options are equally spaced apart. [2]
10. Data security and backup is the responsibility of the researcher. [2]

Question 3: [22]

The entire question 3 is based on the following abstract:

Abstract

Plant diseases affect the growth of their respective species, therefore their early identification is very important. Many Machine Learning (ML) models have been employed for the detection and classification of plant diseases. However, after the advancements in a subset of ML, that is, Deep Learning (DL), this area of research appears to have great potential in terms of increased accuracy. Many developed/modified DL architectures are implemented along with several visualization techniques to detect and classify the symptoms of plant diseases. Moreover, several performance metrics are used for the evaluation of these architectures/techniques. This review provides a comprehensive explanation of DL models used to visualize various plant diseases. In addition, some research gaps are identified from which to obtain greater transparency for detecting diseases in plants, even before their symptoms appear clearly.

1. What does this study aim to achieve? [3]
2. What is the research problem of this study? [3]
3. Explain the methodology used in this study. [4]
4. Identify and explain the evaluation metric used in this study. [2]
5. Does this study need to consider any research ethical clearance? [4]
6. What is the main contribution of this study? [2]
7. Why are there no citations in this abstract? [2]

Question 4 [28]

1. What is the difference between primary sources and secondary sources? [4]
2. What is an in-text referencing? [2]
3. Using an example or illustration, explain the key differences between qualitative and quantitative research. [4]
4. Mention the four most popular tools that are used for creating online surveys [4]
5. Probability sampling is also known as random sampling, the most precise type of sampling. List and briefly describe any two types of probability sampling [4]
6. Explain why gender and marital status are considered nominal data. [2]
7. Explain whether the following data is personal, not personal or sensitive [8]
 - i. Majority of students are Christians
 - ii. Mandisa stays at 23 Smith Street
 - iii. The email address for the company is info@ourcompany.com
 - iv. The political party you are going to vote for on the 29th of May

TOTAL: [100]